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PERSPECTIVE

Confounders, Effect Modifiers, and Mediators: Dealing With "Third Variables" in Cardiovascular Epidemiology

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Research studies in cardiovascular epidemiology assessing the relationship between a primary exposure and a health outcome must account for the presence of what are often called third variables, representing additional characteristics that can affect the association of interest. Even in randomized clinical trials (RCTs), the focus of secondary analysis is often on integrating third variables that can further explain and deepen the understanding of the evaluated treatment effects. In most cases, a third variable operates as a confounder, an effect modifier, or a mediator. The goal of this commentary is to review the definitions of each of these 3 concepts, their interpretation, and the basic analytical tools for their investigation.

CONFOUNDING

In its simplest and most intuitive definition, a confounder is a variable that is associated with both the exposure and the outcome and might cause spurious associations if unaccounted for. For example, a study investigating the association between smoking and the risk of heart failure should account for the participants' level of physical activity, an independent predictor of heart failure that is likely to be associated with smoking status (Figure). Ideally, confounding should be controlled for during study design. In an RCT, for example, the goal of randomization is to prevent third variables of interest from being imbalanced by treatment assignment, thus reducing possible confounding. In a population-based study or in clinical settings where randomization is not feasible or ethical, confounding can be mitigated by multivariable adjustment in regression models or with alternative modern approaches in causal inference.[1]

EFFECT MODIFICATION

An effect modifier is a third variable whose presence or level modifies the primary association of interest. For example, the effect of smoking on heart failure might be enhanced in the subgroup of smokers who also have diabetes (Figure). Effect modification is a question of interest in both epidemiological studies and RCTs and allows the identification of specific subgroups of the population who might be more susceptible to hazards of a specific risk factor or for whom the treatment under examination might be particularly beneficial or harmful. Analytically, effect modification can be addressed by stratification and interaction testing.[2] A critical condition for a robust assessment of effect modification is the level of heterogeneity of the population under study and whether there are enough individuals in the specific subgroups of interest. For example, a study recruiting individuals from private clinics may not be particularly suited for assessing effect modification by social factors such as income or education level if the clinic population consists primarily of those with higher levels of education or socioeconomic status.

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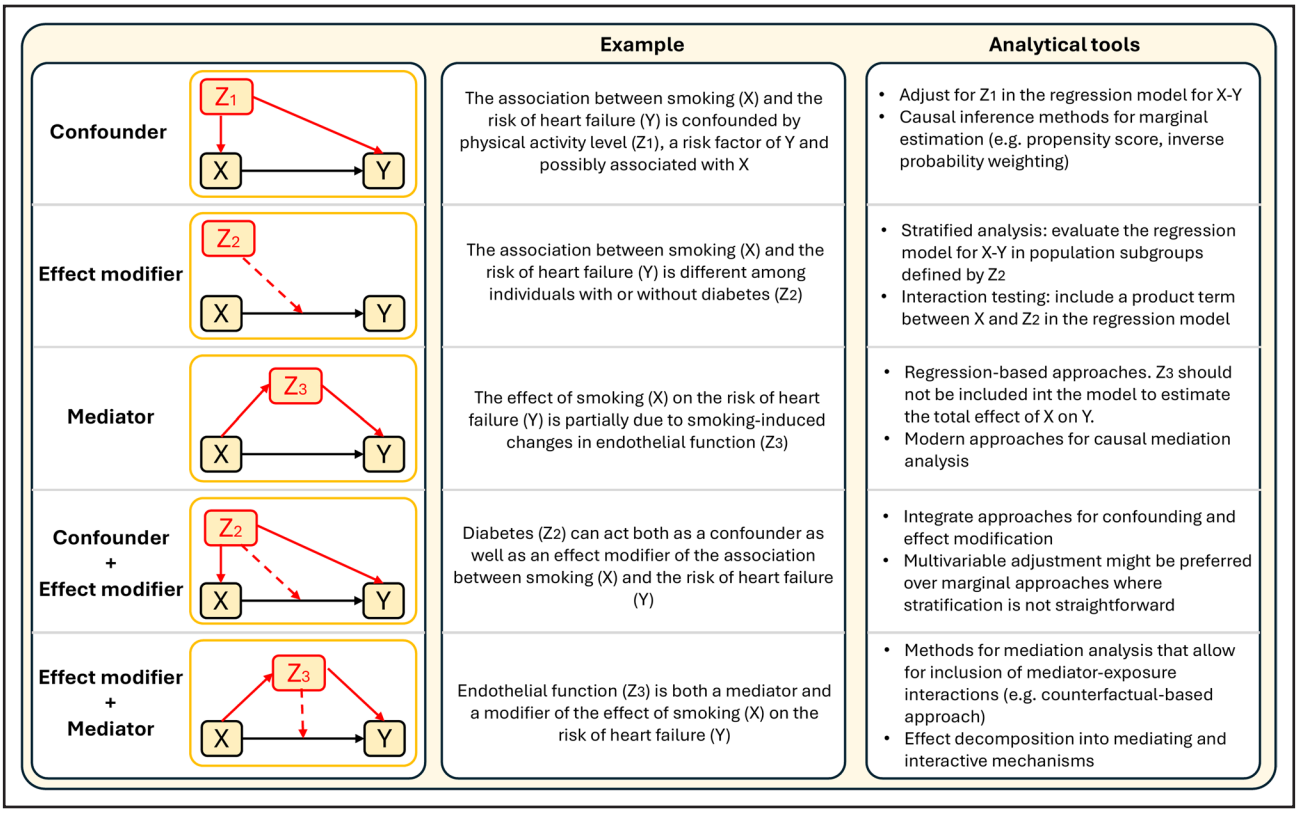

Figure. Illustrations, examples, and overview of analytical tools for assessing third variables in epidemiological studies and secondary analyses of randomized clinical trials.
Third variables generally operate as confounders, effect modifiers, or mediators of the association/effect of interest. A variable could be both a confounder and an effect modifier or a mediator and an effect modifier.

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MEDIATION

A mediator is a third variable that lies on the causal pathway leading from the exposure/treatment to the health outcome and that might partially explain the association of interest. Questions about mediation can be of interest in the secondary analysis of both epidemiological studies and RCTs to study the mechanisms underlying the primary association of interest. Evaluated mechanisms commonly involve biological, psychological, or social pathways.[3,4] For example, a study might investigate the extent to which the effects of smoking on heart failure risk are due to smoking-induced changes in endothelial function (Figure). A formal and thorough assessment of mediation effects requires strong assumptions on the presence of a causal relationship from the exposure to the mediator and the mediator to the outcome.[3] To this end, the mediator should ideally be collected at baseline and again after the treatment/exposure to assess change in the mediator. The development of analytical tools for conducting mediation analysis and strengthening its causal interpretation has been an extensive area of research in modern epidemiology, and several approaches based on regression modeling and alternative techniques have been presented.[5] Of note, even modern approaches will strongly rely on the validity of causal assumptions that cannot be directly tested.

ADDITIONAL SETTINGS

Understanding the differences between confounding, effect modification, and mediation and how they can be incorporated into clinical and epidemiologic studies is critical for correctly conducting an analysis for researchers and for strengthening the interpretation of the results for clinicians. An additional complexity is that, given an exposure-outcome association of interest, a third variable could be both a confounder and an effect modifier or both a mediator and an effect modifier but could not be both a mediator and a confounder.

Confounder and Effect Modifier

The analytical tools for addressing confounding and effect modification can generally be applied to the same dataset. For example, one could adjust for a potential confounder in a regression model to estimate the unconfounded exposure-outcome association and then stratify by that same variable to assess its role as a potential effect modifier.

Effect Modifier and Mediator

This setting has relevant analytical consequences because traditional methods for mediation analysis do not allow the potential mediator to also act as an effect modifier (ie, they do not allow for exposure-mediator interactions). Modern approaches in causal mediation analysis can instead address this scenario and allow disentangling the proportion of total effects due to mediating versus interactive mechanisms.[5]

Confounder Versus Mediator

A variable cannot be both a mediator and a confounder of the same association, and failure to distinguish a mediator from a confounder can have severe consequences. Specifically, mediation analysis requires the potential mediator not to be included in the regression model that estimates the total effect of the exposure on the outcome, which is what would instead be required if that variable were a confounder.

The analytical approaches that we have described provide context for the presence of a third variable. The potential relationship between the different variables should be hypothesized a priori and graphically displayed, and the choice of the analytical tool should follow accordingly (Figure). For observational studies in clinical epidemiology and for the secondary analysis of RCTs, correctly identifying the potential role of each variable of interest is critical to choosing the most appropriate analytical framework, defining the required assumptions, and improving results interpretation and the robustness of scientific findings.

ARTICLE INFORMATION

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